

Lesson 1 Smart Phone App Control

1. Preparation

1.1 Installation

Android user: find the installation package in this folder and install after transferring it to the smart phone. Note: ensure to allow all permissions request for the APP in the phone settings, otherwise it may affect the function implementation.

iOS user: Search “WonderPi” in App Store

Remember to turn on GPS and Wi-Fi before using the app.

1.2 Start ArmPi FPV

1) Place the ARMPI FPV robotic arm on the map correctly according to the folder "1. Getting Ready\ Lesson 2 Preparation."

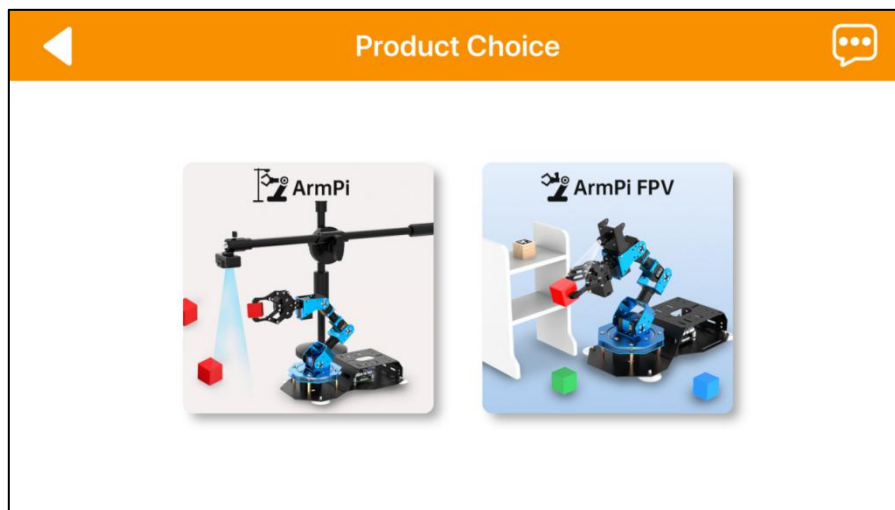
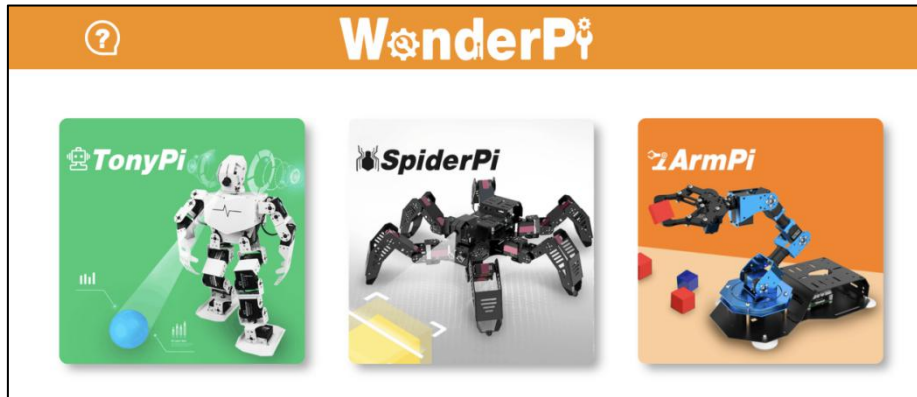
2) Make sure the camera is firmly connected. The camera cable can be connected to any of the 4 USB ports on the Raspberry Pi controller.

3) Turn on Raspberry Pi expansion board and wait to finish.

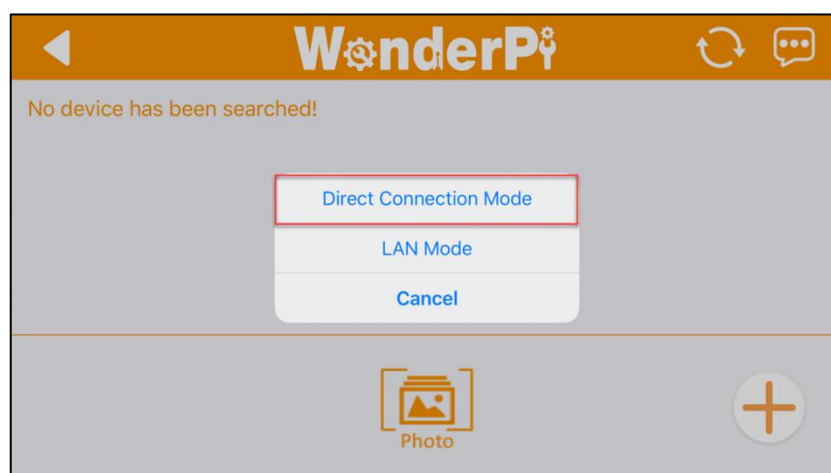
2. Device Connection

After the robotic arm is successfully turned on, it will create a hot spot named with the first letters “HW” by default. By connecting to this hot spot through the APP, you can control the robotic arm now.

- 1) Open the app and select “ArmPi” and “ArmPi FPV” in turn.



- 2) Click “+” button in the bottom right corner and choose “Direct Connection Mode”.



3) Move to phone settings and connect the device hotspot named with the first letter "HW".



i If you are using iPhone to control, you should wait for iPhone to connect the device hotspot successfully, then back to the app. Otherwise, the device ArmPi FPV can not be found on the app and you should click the  to refresh.

4) Back to the app after connecting. The app will connect the ArmPi FPV to your phone automatically. When the robot icon appears, the connection is successful.

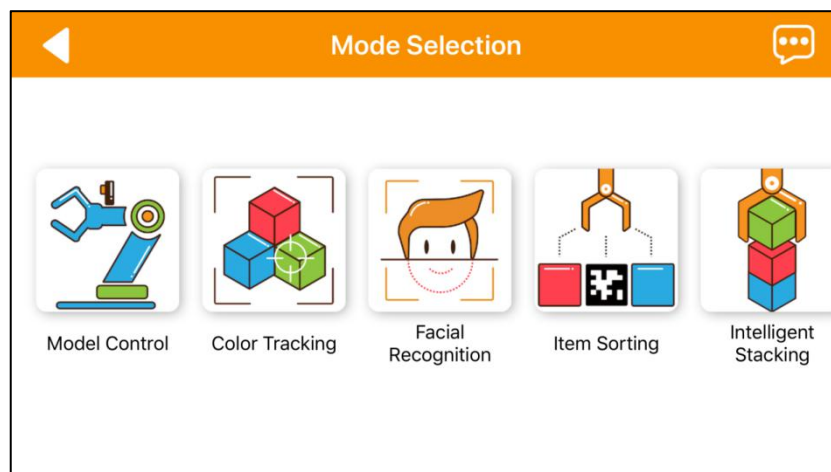


i If you receive pop-up messages showing “whether to keep connecting as the network is unavailable”, just click "keep connecting".

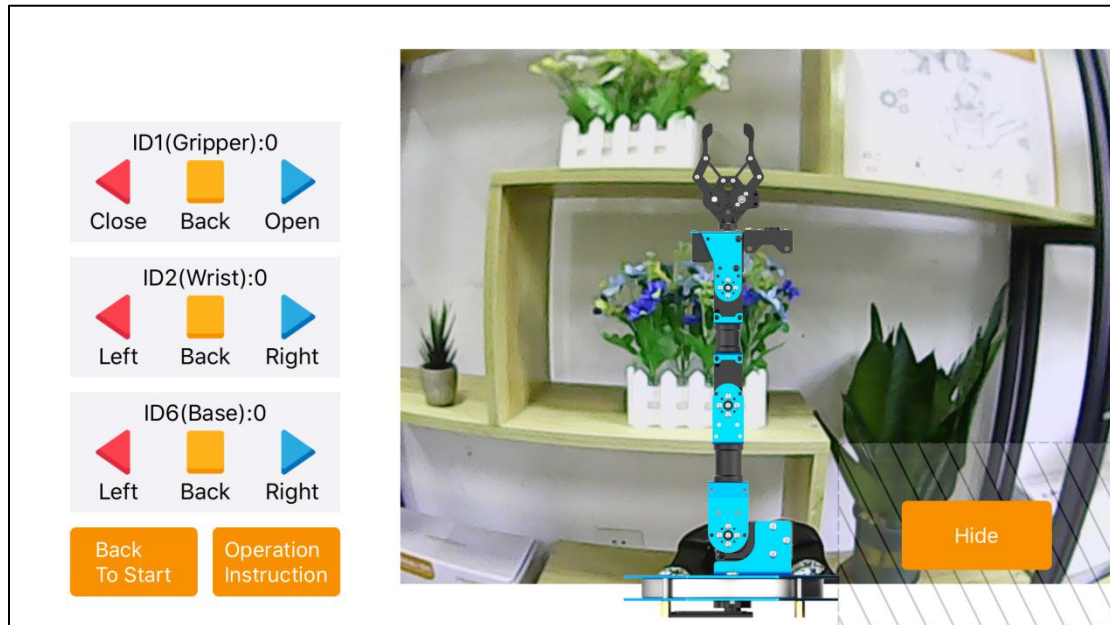
In addition to the methods as shown before, you can connect Raspberry Pi and the phone to the same network at the same time. Refer to “4. Advanced Program Lessons-3.Raspberry Pi Extension Board Tutorial>Lesson 7 methods to connect” for specific instruction.

3. Have a Try

- 1) Click ArmPi to come into the interface.

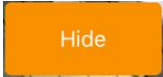


2) There are different games for your choice. When you click any game, the APP will enter the corresponding interface and the robotic arm will return to standard position.

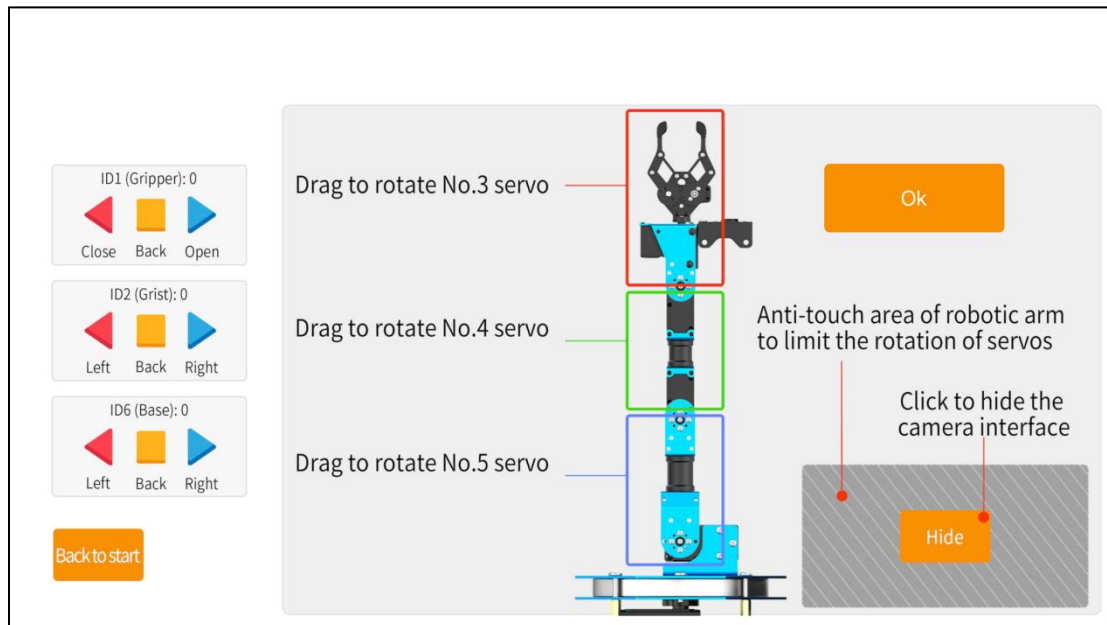


3) The interface of “Model Control” consists of three parts. Buttons on the left sides are used to control the servos of ID2, ID3 and ID6. The right side of the interface is real-time camera interface and a two-dimensional animation. Refer to the table below for corresponding function of the icon.

Icon	Function
ID1(Gripper):0 Close Back Open ID2(Wrist):0 Left Back Right ID6(Base):0 Left Back Right	Click the corresponding icon to control the servo of ID1, ID2 and ID6 respectively, that is, control the opening, closing and rotation of the gripper as well as the movement of the body.
Back To Start	Make all the servos on the ArmPi FPV come to the middle position. (The robotic arm will keep upright.)

	Click "Hide" to hide the camera interface
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4) You can click and refer to the "operation instruction".



Everytime the game is switched, the robotic arm will return to its original posture.

5) If you want to back to the "Mode Selection", please click on blank area, then click  button on the left.

In this chapter, the principle behind the games will not be explained and you can refer to the tutorials in folder "3.AI Vision Games Lesson".

其余玩法各可参考下表:

No.	Game	Outcome
1	Remote Control	Control the movement of the robotic arm

2	Item Tracking	Select the color. Then the robotic arm will move to track the item in selected color
3	Face Recognition	The camera will be lifted, and the robotic arm will perform a set of action when recognizing human face
4	Item Sorting	Select the color and ID of the items to be sorted. And the robotic arm will pick corresponding blocks in recognition area and place them in corresponding sorting area
5	Intelligent Stacking	Pick and stack the blocks in corresponding stacking area after identifying the blocks
6	Intelligent Warehousing	Select the blocks to be warehoused and the shelf for storage. Then the robotic arm will pick the corresponding blocks and place them to the selected shelf.
7	Intelligent Delivering	Select the shelf of the items to be delivered and the place to place. The robotic arm will grab a block from the specified shelf and place it in the corresponding position.
8	Intelligent Transferring	Move the blocks from one side of the shelf to the other side of the shelf, or move the blocks from one layer to other layer on the same shelf

Note: when the page is loading slowly, you can leave the page and enter again.